FAISAL MAHMUD

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SUMMARY

I am a computer science graduate focusing on web privacy and security. For a paper published at PoPETs 2026, I built the crawling pipeline and ran the statistical analysis for the study that used Internet Archive data to trace third-party trackers on government websites in 61 countries back to 1996. My current projects include measuring client-side vulnerabilities on the internet and comparing what agentic crawlers observe against what conventional script-based crawlers see. I see myself pursuing a PhD to go deeper into web privacy and security, highly excited to be pulled into new areas and questions.

EDUCATION

New York University Abu Dhabi — Abu Dhabi, UAE
B.S. in Computer Science, Minor in Applied Mathematics | *GPA: 3.55/4.00* — *Graduated May 2026*

- **Capstone Thesis:** *A Longitudinal Analysis of Privacy Behavior in Android Web Browsers* (2018–2025). Advisor: Dr. Sandra Siby; Secondary Supervisor: Dr. Christina Pöpper.
- **Study Away:** NYU New York (Fall 2024); NYU Paris (Spring 2024).
- **Selected Coursework:** Web Privacy, Computer Security, Machine Learning, Operating Systems, Applied Internet Technology, Computer Networks.
- **Scholarship:** NYUAD full scholarship, 2022–2026.

PUBLICATIONS

Publications

[1] Sachin Kumar Singh, **Faisal Mahmud**, Robert Ricci, and Sandra Siby. "The Empire Strikes Back (at Your Privacy): An Archaeology of Tracking on Government Websites." *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2026(2):108–126, 2026. Presented at the 26th Privacy Enhancing Technologies Symposium (PETS), Calgary, Canada, July 2026. doi:10.56553/popets-2026-0039

RESEARCH EXPERIENCE

Haven Lab, NYU Abu Dhabi — Abu Dhabi, UAE
Research Assistant | *Advisor: Dr. Sandra Siby* — *May 2025 – Present*

- ***Agentic vs. Script-Based Web Crawling*** (current, leading). Measuring whether LLM-driven browser agents observe a different web than conventional script-based crawlers do, by running both against the same live sites and comparing what each one reaches and what each one is served. Instrumented Browser Use v0.13.1 to record agent behavior at the level of individual actions. Exploring security and privacy metrics for the comparison such as tracker exposure, cookie and storage writes, fingerprinting surface, and bot detection and blocking rates.
- ***Longitudinal Client-Side Web Security*** (with S. K. Singh). Built the archival measurement pipeline behind a multi-country study of 26 client-side attack classes, running prototype-pollution, clickjacking, and RetireJS vulnerable-dependency detection over 91,570 Internet Archive snapshots (2003–2024).
- ***Archaeology of Tracking on Government Websites*** (with S. K. Singh). Built the crawling and data-collection pipeline for a study of third-party tracking across 16.5K government websites in 61 countries. Worked through the retrieval problems that make archival data hard to use at scale, including throttled sequential downloading to stay within the Archive's capacity, broken and redirecting URLs, and normalizing Wayback's rewritten resource URLs back to their original form, without which embedded tracker domains cannot be identified at all. Shaped the measurement methodology, including the yearly tracker-labeling rule and the validation against historical EasyList snapshots that justified applying a single 2025 blocklist across three decades, and ran the statistical analysis behind the paper's findings to separate real shifts in tracker adoption from archival noise.

Undergraduate Capstone Project — Abu Dhabi, UAE
Independent Researcher | *Advisor: Dr. Sandra Siby* — *Aug 2025 – Dec 2025*

- Built an automated Android testbed to measure how five mobile browsers (Brave, DuckDuckGo, Firefox, Kiwi, Opera) changed their default privacy protections between 2018 and 2025, installing and exercising 39 archived APKs on a controlled Pixel 3a emulator running Android 10.
- Designed five in-browser probes covering third-party cookie blocking, SameSite enforcement, storage API isolation, tracker-domain blocking against a Disconnect-derived list, and DNS-over-HTTPS detection, driven by a Python multi-origin test server over ADB reverse port forwarding with all session output logged to structured JSON.
- Found that cookie and SameSite protections have converged across every browser and year, leaving tracker blocking as the only dimension that separates them: Brave climbed from roughly 10% to 60% blocking over the period while DuckDuckGo and Kiwi blocked nothing at the network level, and no browser enabled DNS-over-HTTPS by default in any year tested.

Cyber Security and Privacy Lab, NYU Abu Dhabi Abu Dhabi, UAE
Research Assistant | Advisor: Dr. Christina Pöpper *Jun 2024 – Jul 2024*

- Analyzed spoofing and injection attack surfaces in the ADS-B aviation surveillance protocol, applying systems-security threat modeling to unauthenticated broadcast messages.
- Implemented Python and MATLAB tooling to decode and validate captured ADS-B message logs, and built the processing pipeline used to flag anomalous aircraft tracks in the dataset.

Talks & Posters

Poster: "The Empire Strikes Back (at Your Privacy): An Archaeology of Tracking on Government Websites." 26th Privacy Enhancing Technologies Symposium (PETS), Calgary, Canada. Jul 2026

Honors & Awards

PETS 2026 Student Travel Stipend, Privacy Enhancing Technologies Symposium 2026
Runner-Up, MENA Region, CSAW'25 LLM CTF Attack Competition, NYU Abu Dhabi 2025

- *CSAW'25:* Built an autonomous CTF-solving pipeline on top of an agentic LLM harness, with multi-step prompting strategies for problem decomposition, tool invocation, and iterative exploit development across pwn, web, reverse-engineering, and forensics categories.

Technical Skills

Languages: Python, JavaScript, Go, C, MATLAB, Bash, LaTeX
Measurement & Instrumentation: Playwright/Puppeteer, `mitmproxy`, Frida, Browser Use, Internet Archive Wayback APIs, OpenWPM-style crawling, Docker, Linux
Analysis: pandas, NumPy, scikit-learn, statistical modeling of longitudinal data
Security Tooling: RetireJS, prototype-pollution and clickjacking detection, TLS interception, ADS-B decoding
Languages (spoken): Bengali (native), English (fluent).

References

1. Dr. Sandra Siby
Assistant Professor of Computer Engineering
New York University Abu Dhabi
Email: sandra.siby@nyu.edu
Relationship: Research advisor (Haven Lab), capstone advisor, and co-author

2. Dr. Robert Ricci
Research Professor, Flux Research Group
Kahlert School of Computing
University of Utah
Email: ricci@cs.utah.edu
Relationship: Research advisor, and co-author, PoPETs 2026